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Toxicology of natural and synthetic aphrodisiacs

Introduction

Aphrodisiacs have been used by several ancient cultures without knowledge of their scientific way of action (aphrodisiac, Greek for sexual pleasure) [1]. Today, aphrodisiacs are generally regarded as substances that increase sexual desire (i.e. libido) and/or sexual pleasure. The definition has been extended to include those substances which enhance sexual performance or aid in the proper functioning of the male and female sex organs [2, 3]. They can be classified according to their effects when consumed or administered. Aphrodisiacs can have psychological effects, thereby increasing sexual desire and pleasure through hallucinogenic properties or other mood stimulating properties; however, aphrodisiacs can also act physiologically, enhancing erection through hormonal changes, increased blood flow, and smooth muscle relaxing properties [2].

Substances used as aphrodisiacs include synthetic chemicals, herbal and animal products and foods, beverages, vitamins and trace elements [4]. The use of these supplements to enhance erectile function as well as sexual arousal and desire has become increasingly popular in recent years and the availability on the internet is increasing [5, 6]. The annual sales of these so-called dietary supplements have doubled between 1999 and 2007 and a survey in the USA showed that over 50% of respondents reported using products for the improvement of male sexual health [7]. It has been estimated that approximately six million men in the EU purchase these products from the Internet [8]. The products are advertised on the Internet as “natural” and “safer” alternatives to pharmaceutical products, such

as the gold standard for treatment of erectile dysfunction (ED) phosphodiesterase type 5 (PDE-5) inhibitors (e.g. sildenafil, tadalafil, and vardenafil); however, they often contain potent active constituents which are undeclared on the packaging [9]. Gilard et al. [10] analyzed 150 herbal dietary supplements marketed for improving sexual performance. Only 31% of the samples could be considered as true herbal or natural products. Furthermore, with reduced medical control and control of drug registration agencies, there are just a few resources for consumers to verify the ability of these products to deliver the advertised effects or to investigate potential side effects. Since herbal supplements are plant products, they are generally perceived and hence marketed as natural and, therefore, “safe”; however, several case reports of side effects from herbs and herbal products have been reported [11], such as on-target and off-target effects from biologically active ingredients, side effects caused by contaminants, and herb-drug interactions. It is difficult to determine the true frequency of side effects from herbs, herbal products and synthetic products sold on the Internet, as surveillance systems are much less extensive than those in place for pharmaceutical drugs.

The phenomenon of free availability of these products combined with the simulated absence of side effects represents a challenge from a forensic toxicological perspective. Synthetic as well as natural aphrodisiacs could potentially be used in drug-facilitated sexual assaults to increase libido of victims or to increase sexual potency to facilitate sexual contact. Both females and males could be victims of drugging. In this review a case report is described in which this scenario was

reported to have happened by the defendant. Furthermore, an internet search and a literature search were started to find the most advertised and top selling products and their potential toxicity and side effects. Scientific studies on the potency of these substances as an aphrodisiac which are relatively rare are not part of this review and are summarized, e.g. by Melnyk and Marcone [12].

Case report

A 54-year-old man had invited several women on several occasions with the aim of having sex with them. Some of the women described similar symptoms after drinking concoctions prepared by the defendant. These symptoms included disorientation, nausea and vomiting, dizziness, problems in vision and speaking, numbness, psychomotor problems, feeling not to be aware of their senses, disinhibition, euphoria and anterograde amnesia. Naked photos of two of these women were found on his computer which showed the women in unnatural sleeping positions obviously draped to shoot sexual photographs. All of the women described the beginning of the symptoms 15–30 min after the intake of the drink and a duration of action of 2–6 h. These symptoms and pharmacokinetic aspects indicated the administration of a knockout drug (especially gamma-hydroxybutyric acid or its precursors); however, the defendant claimed that he had ordered a liquid on the internet that was supposed to have aphrodisiac effects on female and male users. He claimed that he wanted to ease sexual contact with these women and that he wanted to improve their sexual feelings. He claimed that he did not know which

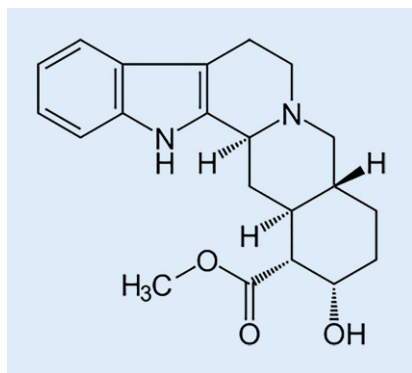


Fig. 1 ▲ Chemical structure of yohimbine

ingredients were in the product. A joint search for the product on the internet was unsuccessful. Furthermore, forensic samples of the victims could not be analyzed.

Aphrodisiacs with sexually enhancing properties and the side effects

Synthetic substances

Phosphodiesterase V inhibitors

Recently synthetic substances, such as sildenafil or other phosphodiesterase 5 (PDE-5) inhibitors are taken to treat ED. These are supposed to only work when the patient is sexually stimulated; however, the substances themselves do not increase libido [3]. Therefore, they only work in male users. In Germany, products containing sildenafil (Viagra® 25 mg/50 mg/100 mg), tadalafil (Cialis® 5 mg/10 mg/20 mg), vardenafil (Levitra® tablets 5 mg/10 mg/20 mg) are officially available. Furthermore, to date more than 50 analogues of PDE-5 inhibitors have been reported [13]. Gilard et al. [10] analyzed 150 herbal dietary supplements marketed for improving sexual performance. Although all of these formulations were claimed to contain only natural compounds, plant extracts and/or vitamins, 61% of the products were adulterated with phosphodiesterase-5 inhibitors (27% with sildenafil, tadalafil and vardenafil, and 34% with structurally modified analogues). Among them, 64% contained only 1 PDE-5 inhibitors and 36% mixtures of 2, 3 and even 4. The amounts of PDE-5 inhibitors

medicines were higher than the maximum recommended dose in 25% of dietary supplement tainted with these drugs [10]. The most common adverse events are due to PDE-5 inhibition in vascular and visceral smooth muscle and are mainly cardiac effects; however, as these substances are part of several reviews on side effects on sexual enhancing drugs (e.g. [14]) and intoxications have been described (i.e. [15]), this will not be further described in detail here. Herbal phosphodiesterase inhibitors, a future perspective of new drugs, are reviewed by Rahimi et al. [16].

Yohimbine

Yohimbine (see structure in ■ Fig. 1) is an indole alkaloid from the leaves and the bark of the yohimbe tree, *Pausinystalia johimbe*, that has important forensic implications due to its varied applications [17]. A study by Betz et al. on dietary supplements containing yohimbe extracts has shown that the yohimbine content may range from <0.1 to 489 ppm compared with 7089 ppm in the authentic plant material, which has been attributed to the intense dilution of final products [18]. Modern yohimbine manufacturing and purification for pill formation is achieved through ethanol or chloroform extraction. The yohimbe bark is ground and submerged in the solvent, followed by filtration and evaporation of the solvent [19].

It is a highly potent antagonist on presynaptic and postsynaptic α_2 -adrenoreceptors within the smooth muscles and blood vessels. The enhancement of sexual impulse is generally believed by blocking the α_2 -adrenoreceptors in the locus ceruleus in the brain [20], which mediates erection-inhibiting impulses in the central nervous system [21]. Yohimbine has also been abused on the streets as an aphrodisiac and hallucinogen, while more recently it has gained popularity in the body building community for its lipolytic and sympathomimetic effects for fast weight loss and body building supplement [17]. Because of the resulting dilatation of the vessels, it has been approved for use in the USA and in Germany (Yocon Glenwood tablets® 5 mg) to treat ED. An average oral dose

of 5–15 mg (1–3 tablets) is taken for the treatment of ED [22]. Due to its central and libido-enhancing effects, however, if given in drug-facilitated sexual assaults, not also men but also women could be the victims of yohimbine administration.

The side effects of yohimbine include high blood pressure, increased heart rate, manic reactions, bronchospasm, palpitations, insomnia, anxiety, irritability, shivering, sweating, nausea, flushing, and headaches, which can all be attributed to its central adrenergic activity [23]. According to the California poison control system, 238 cases of adverse reactions after yohimbine consumption have been identified within a 7-year period (2000–2006), where the most commonly reported adverse events were gastrointestinal distress, tachycardia, anxiety, agitation, and hypertension [24]. Overdoses leading to neurotoxic effects were seen from doses of 200–5000 mg [17]. Symptoms reported with yohimbine overdose include anxiety, drowsiness, disorientation, tremors and seizures with higher doses [17]. Furthermore, yohimbine intake in a fasting state can lead to panic attacks [25]. Some case reports after yohimbine intoxication are described reporting of side effects, such as anxiety, tremors, hypertension [26], nausea, headache, rash and dissociative reactions [27] drowsiness, confusion, atrial fibrillation, seizures [17, 28] and unconsciousness [29]. Interestingly, Varkey et al. [30] additionally reported retrograde amnesia after the intake of 350 mg in a 38-year-old man. Sandler and Aronson described the development of progressive renal failure, cutaneous eruption, and a lupus-like syndrome in a 42-year-old Afro-American man following yohimbine use [31]. Myers and Barrueto reported a case of refractory priapism in a 42-year-old man who self-administered a product containing yohimbine to treat ED and was subsequently admitted to emergency department with a 20-h lasting erection [32]. In 1985 Linden et al. reported a case of a 16-year-old girl who experienced an acute dissociative reaction accompanied by weakness, paresthesia, incoordination, anxiety, headache, and chest pain after the ingestion of

an aphrodisiac called “yo-yo,” which has been later identified as yohimbine [27]. Acute neurotoxic effects related to the ingestion of yohimbine-containing products were reported by Giampreti et al. [17] in the case of a 37-year-old bodybuilder who presented with malaise, vomiting, loss of consciousness, and repeated seizures after ingestion of 5 g of yohimbine during a competition [17].

Rauwolfscine (alpha-yohimbine) is an alkaloid in *Rauwolfia* species and a stereoisomer of yohimbine with comparable aphrodisiac properties [33].

Steroid hormones

Libido is affected by several hormones including testosterone [34]. In postmenopausal women and in older men with decreased libido, which correlates with decreased testosterone concentration, steroid supplements can increase libido [35]. Other steroids which can imitate the action of testosterone can also enhance libido, i.e. tibolone which is a synthetic steroid hormone that can increase lust, agitation and the capability to have an orgasm [36]; however, in libido enhancing products sold on the internet mostly dehydroepiandrosterone (DHEA, prasterone, chemical structure see ■ Fig. 2) is present [37], which is an endogenous substance produced in the adrenal cortex and the precursor for female estrogens and male androgens. In Gynodian® ampules DHEA is used to treat symptoms of the menopause in women. Furthermore, anti-aging products from the internet contain DHEA [38]. The following side effects have been described after the intake of Gynodian®: icterus, hypertension, headache/migraine, somnolence, dizziness, swelling and redness of the legs, sudden chest pain and breathlessness. No major adverse effects were reported during a study period of 16 weeks in women who received 25 mg DHEA 3 times a day. As expected in steroid users, up to 22% of subjects in the DHEA group complained of increased acne compared with 8.3% in the placebo group, and up to 22% in the DHEA group and 16.7% in placebo group had a nonpersisting mild elevation of liver enzymes [39].

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Toxicology of natural and synthetic aphrodisiacs

Abstract

Aphrodisiacs, libido or erection-enhancing supplements are widely available on the internet. The beneficial effects are controversial. Forensic toxicologists should be aware of the potential toxicities of these products because they could be given in cases of drug-facilitated sexual assault. This article presents a case report in which the defendant claimed to have administered an aphrodisiac to several women who showed strong symptoms of intoxication after consumption in an alcoholic drink. A literature research was carried out for potential substances sold in these aphrodisiac products. Synthetic substances as well as products of herbal or animal origin can be the

ingredients. The main products that could cause intensive side effects and intoxication are phosphodiesterase V inhibitors (declared or undeclared on packaging), yohimbine, steroid hormones, flibanserin, gamma-hydroxybutyric acid, *Tribulus terrestris*, Bufo Toad, Spanish fly or Horny Goat Weed. All other products used as aphrodisiacs and described herein show less potential side effects or no studies on side effects in humans were carried out.

Keywords

Aphrodisiac · Toxicology · Forensics · Yohimbine · *Tribulus terrestris*

Toxikologie natürlicher und synthetischer Aphrodisiaka

Zusammenfassung

Aphrodisiaka, die Libido oder die Erektion steigernde Ergänzungsmittel sind im Internet weit verbreitet und leicht erhältlich. Ob diese positive Effekte haben, ist bei einigen dieser Substanzen zumindest fragwürdig. Forensischen Toxikologen sollte die potenzielle Toxizität dieser Produkte bekannt sein, denn diese könnten in Drug Facilitated Sexual Assaults verabreicht werden. In diesem Beitrag wird ein Fall präsentiert, in welchem der Beschuldigte behauptete, mehreren Frauen ein im Internet gekauftes Aphrodisiakum verabreicht zu haben, nach dessen Konsum die Frauen heftige Intoxikationssymptome aufwiesen. Es wurde eine Literatursuche nach potentiell in diesen Produkten enthaltenen Substanzen durchgeführt. Synthetische Substanzen, natürliche Produkte, aber auch Substanzen

tierischen Ursprungs können Inhaltsstoffe sein. Diejenigen Produkte, die am ehesten zu Nebenwirkungen oder Intoxikationen führen können, sind Phosphodiesterase-V-Inhibitoren, Yohimbin, Steroidhormone, Flibanserin, Gamma-Hydroxybuttersäure, *Tribulus terrestris*, Kröteninhaltsstoffe, die Spanische Fliege oder Horny Goat Weed (Epimedium). Alle weiteren in diesem Review beschriebenen Produkte, welche als Aphrodisiaka genutzt werden, zeigen weniger Nebenwirkungen bzw. es wurden hierzu keine Studien zu Nebenwirkungen im Menschen durchgeführt.

Schlüsselwörter

Aphrodisiakum · Toxikologie · Forensik · Yohimbin · *Tribulus terrestris*

Flibanserin

Flibanserin (chemical structure see ■ Fig. 3) is a drug which was approved by the Food and Drug Association (FDA) (Addyi®) for the treatment of hypoactive sexual desire disorder (HSDD) in women. Flibanserin was identified in 2 of 150 supplements from the internet marketed for improving female libido [10] and low levels were identified (11 mg per capsule) in a pharmacy-sold herbal dietary supplement for female HSDD treatment [40]. A clinical trial for

the substance demonstrated its efficacy against HSDD for women treated with a 100 mg daily dose for 24 weeks [41].

Flibanserin is an agonist of 5-HT_{1A} and an antagonist of 5-HT_{2A} receptors, furthermore a partial agonist on D₄ receptors [42]. The most frequently reported adverse events in the flibanserin group during a clinical study were somnolence, fatigue, dizziness, and nausea, with adverse events leading to discontinuation in 9.6% of women receiving flibanserin vs. 3.7% on placebo [41].

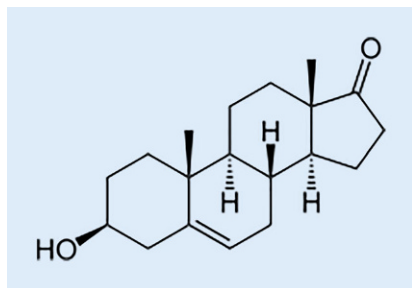


Fig. 2 ▲ Chemical structure of DHEA

Further documented occasional side effects were abdominal pain, insomnia, palpitations and obstipation [41]. Safety assessments included the proportion of women who experienced an adverse event (odds ratio OR 1.54, 95% confidence interval, CI 0.34–1.76, $P < 0.00001$), nervous system disorders and fatigue indicated that flibanserin was generally well tolerated [43]. In another study, no flibanserin-related serious adverse effects and no instances of suicidal ideation were seen in 595 patients [44]; however, Granzella et al. [45] described a case of a 2-year-old boy who ingested an estimated 600 mg of his mother's flibanserin. Following ingestion, the child developed facial twitching and unresponsiveness to pain. In the emergency department he was found to have hypertension, mydriasis and slurred speech. He responded well to supportive care including administration of benzodiazepines. Shortly after admission to the hospital, his temperature increased to 38.4 °C. Interestingly for forensic toxicologists, flibanserin is metabolized to 1-(3-trifluoromethylphenyl)-piperazine (TFMPP), a recreational drug used as an alternative for MDMA (Methylenedioxy-methamphetamine) in ecstasy tablets [45].

Arginine

In many countries, including Germany, dietary supplements and functional foods containing amino acids are available for non-prescription sales. One of these amino acids is arginine (chemical structure see ■ Fig. 4), a semi-essential amino acid involved in many parts of the human physiology and metabolism which became increasingly more popular [46]. It is involved in ammonia detoxification,

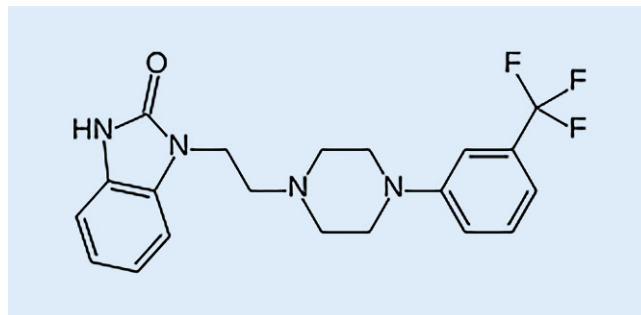


Fig. 3 ◀ Chemical structure of flibanserin

due to its role as an intermediate in the urea cycle, hormone secretion and immune modulation [47]. Containing four nitrogen atoms per molecule, arginine is also well known as a biologic precursor of nitric oxide, which is an endogenous messenger molecule playing a big role as an endothelial relaxing factor leading to various roles in endothelium-dependent physiological effects in the cardiovascular system [47, 48]. Many of the potential clinical effects of arginine are thought to be mediated by its NO-stimulating effects [47]. One of the referred therapeutic applications is ED as well as male and female infertility [47, 48].

The observed safe level (OSL) for arginine, based on the available published human clinical trial data and indicating evidence for the absence of adverse effects, is 20 g per day for normal healthy adults. Data for intakes above this level are not sufficient. The highest oral dosage for arginine used was 42 g per day for 6 weeks with no adverse effects reported [46]. Overall significant adverse effects due to arginine supplementation have not been observed in studies on animals or humans [46–49]. Nevertheless, it must be noted that people with renal failure or hepatic diseases might be unable to metabolize and excrete arginine to its full extent [47]. Furthermore arginine, lysine and histidine, all basic amino acids, share the same transport system on the plasma membrane, so they can affect the absorption of each other leading to an amino acid imbalance, which could cause severe adverse effects and even death [49]. Due to this Wu et al. estimated that a 70 kg human subject should be able to tolerate 6 to 15 g per day arginine in addition to a basal amount of arginine from regular food [49].

Some published studies reported side effects, such as nausea, gastrointestinal discomfort and diarrhea possibly due to a rapid and excess production of NO [46, 49, 50]. Grimble reviewed that large single doses seem to provoke diarrhea [50]. A “Risk assessment for the amino acids taurine, L-glutamine and L-arginine” by Shao and Hathcock suggested that there are no dose responsive gastrointestinal effects as well as any other adverse events using equal, higher or lower doses as used in the few studies which showed adverse effects. Findings of adverse events in studies and case reports were: hypotension, tumor stimulation, acidosis, hyperkalemia and cardiac arrest [46, 47, 50].

Phenethylamines

Among synthetic substances used as sexual enhancers, stimulating and aphrodisiac phenethylamines, such as amphetamines, MDMA (ecstasy), cocaine, ephedrine and phenolamine, a synthetic oral drug for boosting sexual performance in man which was detected in a single of 150 formulations at a dose of 7 mg per day of powder [10] and cathinone derivatives like methylenedioxypyrovalerone (MDPV) or 4-methylmethcathinone (mephedrone) or other substances like 4-bromo-2,5-dimethoxyphenylethylamine (2C-B) were described [51]. Side effects of amphetamine derivatives are well known and summarized in various reviews (i.e. [52]).

Gamma-hydroxybutyric acid (GHB) and its precursors

Gamma-hydroxybutyric acid (GHB, chemical structure see ■ Fig. 5) is produced endogenously by metabolism of the neurotransmitter gamma-aminobu-

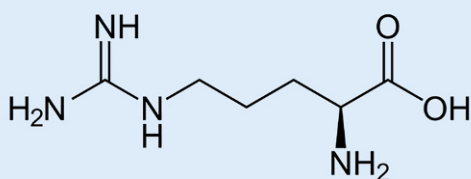


Fig. 4 ◀ Chemical structure of arginine

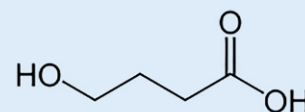


Fig. 5 ▲ Chemical structure of GHB

tyric acid (GABA) and usually given as a knockout drug during drug-facilitated crimes due to its narcotic effects; however, GHB is also an aphrodisiac in man. It has four sexually enhancing effects: disinhibition (e.g. relaxation), heightened sense of touch, enhancement of male erectile capacity and increased power of orgasm [53, 54]. Adverse effects are dose-dependent and are part of several reviews. Briefly, GHB doses of up to 1.5 g lead to a euphoric status, relaxation and disinhibition. It acts as an anxiolytic and has socially opening effects comparable to a mild alcohol intoxication. In doses up to 2.5 g mood and impulse are increased and GHB acts as an aphrodisiac. The narcotic effects are primarily seen with higher doses >3–4 g and GHB shows a stimulating-sedative profile: within the first hours the psychostimulating effects dominate [55]. Side effects are described 15 min after the intake and include headache, somnolence, psychomotor difficulties, nausea and vomiting, dizziness, hypertension, disturbances of speech, seizures, disorientation, and can lead up to drowsiness, deep sleep, unconsciousness and death due to breathing arrest [55, 56]. Duration of action of GHB can be 1–7 h depending on the dose [57]. Precursors of GHB which can be converted into GHB within the body and are easier available (gamma-butyrolactone and 1,4-butanediol) can also be taken as aphrodisiacs.

Products with herbal or animal origin

Tribulus terrestris and other protodioscin-containing species (fenugreek and *Dioscorea*)

Tribulus terrestris (TT) is an annual herb found in temperate climates all over the

world [3]. The plant, its fruits and its extracts have traditionally been used as a sexual stimulant [58]. According to *The Canon of Medicine* and Aghili Khorasani's *Makhzan al-Advia* (The treasury of spices, eighteenth century), Bindii or TT influences libido [58] and is able to boost sex drive in human beings; however, it is also supposed to have a positive effect on penile erection [59]. It is most often used for infertility and loss of libido, therefore it is also imaginable to drug women and men. It is also present in over the counter herbal supplements purportedly able to treat various ailments such as urolithiasis [60], cardiovascular disease [61], and for the use of enhancing physical performance of body builders [62].

The aphrodisiac properties of TT are due to the steroidal saponin protodioscin (PTN, chemical structure in ■ Fig. 6), which is also ingredient of the fenugreek (*Trigonella foenum graecum*) [63] and *Dioscorea* (yam) species [64]. Both have been mentioned to be used as aphrodisiacs in the literature.

Regarding potential side effects, on internet pages selling these products mostly gastrointestinal problems and icterus are listed. In a randomized double-blind placebo-controlled study using TT for treatment of sexual dysfunction, women took 7.5 mL of a 60% solution of the drug over the course of 4 weeks twice daily. Women described side effects including abdominal pain, cramp, nausea, vomiting, diarrhea and constipation [65]. Toxicity in animals is also well documented. A TT extract showed central nervous system (CNS) stimulant activity at a dosage of 20 mg/kg in rats [66]. The TT toxin causes progressive astrocyte degeneration, glutamatergic neuronal excitation reduction and decreases subthalamic nucleus normal

excitatory function [67]. Motor neuron disease [68], neuromuscular ataxia, hepatotoxicity, and photosensitization termed geeldikkop [69, 70] have been described in animals in the natural and in experimental settings. In humans, mainly nephrotoxicity was described [71]. Ryan et al. [72] described a case of TT intoxication in a young healthy male who consumed the drug (one tablet a day for a few months) for body building reasons, presenting with severe hyperbilirubinemia followed by acute renal failure and bile containing casts in the tubules. A similar case was described by Talasaz et al. [73] with neurotoxicity, hepatotoxicity, and renal toxicity in a young healthy male who consumed large quantities of TT extract, for its antiurolithic properties [73]. Serum creatinine rose up to 17 mg/dL, requiring dialysis, and urinalysis findings were suggestive of acute tubular necrosis (ATN). Additionally, the patient developed hypertension, seizures, and markedly elevated serum aminotransferases. Necropsy findings included distortion of bile ducts with yellow-green birefringent crystals, inflammation, apoptotic hepatocytes, and cholestasis [72]. The hepatotoxin is believed to be a further steroidal saponin called diosgenin [70]. These toxic side effects have, however, only been described with frequent applications.

Ambrein

Ambergris is a physiological product found either in the intestines of the sperm whale, *Physeter macrocephalus* L. (Physeteridae) and is mostly discarded into the sea as a waste product (emesis or feces). It occurs in gray or brown waxy masses and consists primarily of undigested squid and cuttlefish beaks, the sperm whale's main food, mixed with fatty digestive fluids [74]. The triterpene

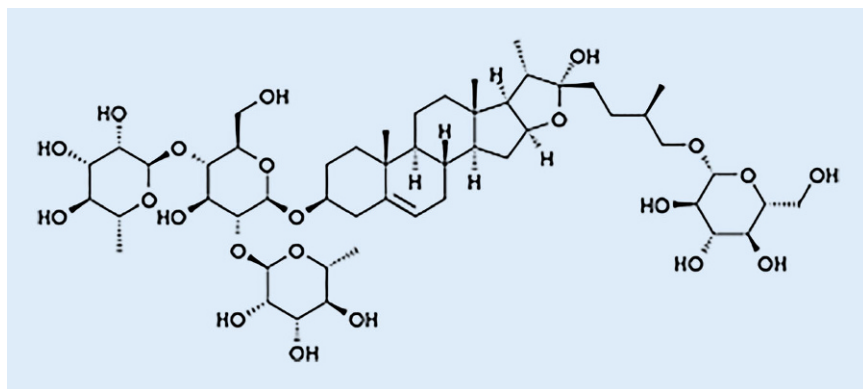


Fig. 6 ▲ Chemical structure of protodioscin

alcohol ambrein (structure see ■ Fig. 7) was isolated as the major constituent of *Ambergris* [75]. Ambrein capsules are sold over the internet. Historically, its use ranged from perfume preparations to the treatment of health conditions. It has been used to enhance sexual performance, with studies supporting its validity as a natural aphrodisiac in male rats [76]. The intake led to a general depressed activity and somnolence in mice with a LD_{50} described to be 7500 mg/kg body weight after intraperitoneal intake. A subcutaneous dose of 250 mg/kg of ambrein markedly decreased both ambulatory activity and cerebral contents of norepinephrine, dopamine and their metabolites in mice [77]. Side effects in humans have not been described in the literature or on the internet.

Bufo Toad (love stone)

The skin and gland secretions of several *Bufo* species have traditionally been used for medicinal and aphrodisiac purposes [78]. The ancient Chinese medication Chan su and Indian aphrodisiac love stone are produced from these secretions [79]. If used as an aphrodisiac, Bufo toad is administered topically. Its aphrodisiac properties is explained by the presence of the psychoactive substances bufotenine and its O-methylated derivative 5-methoxy-N,N-dimethyltryptamine (5-MeO-DMT) (chemical structure see ■ Fig. 8), which are hallucinogenics that provide stimulatory effects [2]. Ayahuasca preparations can also contain 5-MeO-DMT [80]. Both substances are structurally related with

serotonin and psilocin and have a high affinity to the 5-HT_{1A} receptor.

These hallucinogenic properties are the reason why Bufo Toad is only seldomly used as aphrodisiac. 5-MeO-DMT is a potent, fast-acting hallucinogen with short duration of action in humans. Human self-experiments have revealed that 5-MeO-DMT causes visionary and auditory changes and distorts the perception of time. The effects start at 3–4 min, peak after about 35–40 min, and end around 60–70 min after insufflation [81]. Studies with mouse, rat, sheep, and monkey models revealed remarkable ataxia, mydriasis, head nodding, tremor, convulsion and shivering after administration of 5-MeO-DMT. Sheep were found to be the most susceptible to 5-MeO-DMT and exhibited stringy salivation, tachycardia, and respiratory failure when exposed to 1 mg/kg of 5-MeO-DMT. The LD_{50} values of 5-MeO-DMT in mice ranged from 48 mg/kg to 278 mg/kg for different administration routes [82, 83]. In humans, optical hallucinations, such as flashes, dizziness, nausea, hypertension and disorientation have been described.

During 1993–1995 the New York City Poison Control Center (NYCPCC) was informed about onset of illness in five previously healthy men after they ingested a substance marketed as a topical aphrodisiac; four of the men died [84, 85]. They all took Chan Su—used as a topical anesthetic and cardiac medication—which contained bufotenin. Chan su also contains bufadienolides, steroids which inhibit the Na⁺/K⁺-ATPase. In New York, the product was marketed

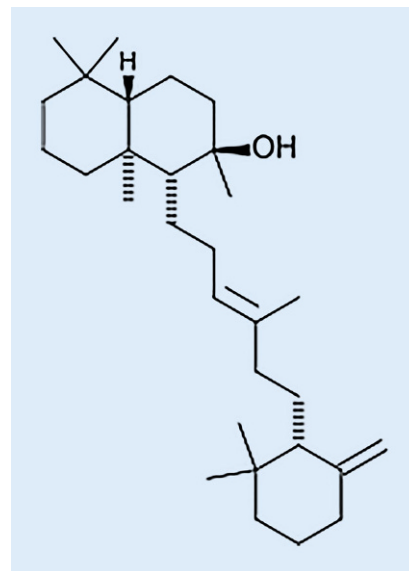


Fig. 7 ▲ Chemical structure of ambrein

as an aphrodisiac and sold under trade names, such as “Stone,” “LoveStone,” “Black Stone,” “Piedra China,” “Jamaican Stone,” “China Rock” and “Rock Hard” and was available in grocery stores and smoke shops and from street vendors at that time. The stone substance is hard, dark brown, and sold as a solid chunk; however, similar cases happened in New York later on [86] and in 2017 cases of “death by love stone” were still described [85]. Lopez Castillo et al. described a 38-year-old patient who suddenly developed runs of complete AV block (atrioventricular) with narrow QRS complex alternating with runs of fast sustained ventricular tachycardia [85]. Typical presentation after the intake of Love Stone also includes acute abdominal pain and bradyarrhythmia in combination with tachyarrhythmia, including atrial, junctional and ventricular tachycardia. Bidirectional and fascicular tachycardia are very specific findings. Studies have demonstrated the negative effects of Chan su and Bufo toad secretions on heart rate and cardiomyocyte contractility by blocking Na⁺/K⁺-ATPase but have not related any findings to increased or decreased sexual performance [78, 79].

Spanish fly (Cantharides)

The Spanish fly is the most well-known of a collection (more than 2000 species)

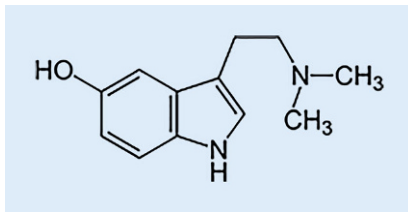


Fig. 8 ▲ Chemical structure of bufotenin. Its O-methylated derivative 5MeO-DMT is also a component of Bufo toad

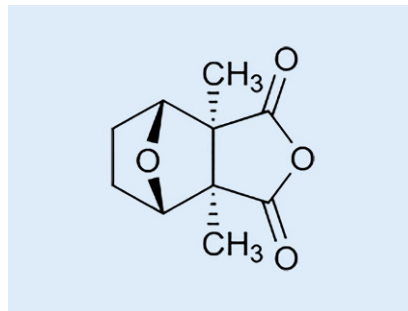


Fig. 9 ▲ Chemical structure of cantharidin

of beetles traditionally used as aphrodisiacs. All of the beetles contain a vesicant liquid and are known as blister beetles. The beetles are most commonly found in southern Europe, Africa, and Asia. The Mediterranean beetle *Cantharis vesicatoria* is the most well known as the Spanish fly, while *Epicauta vittata* is the most common of over 200 blister beetles identified in the USA [87]. The beetle secretes cantharidin (structure see ■ Fig. 9), a toxin used as a part of its defence mechanism and found in both larval and adult forms of the beetle. When agitated, larvae secrete cantharidin as a milky oral exudate, while adults release it from leg joints.

Most commonly available preparations of Spanish fly contain cantharidin in negligible amounts [88]; however, the chemical is available illicitly in concentrations capable of causing severe toxicity. At high doses cantharidin is toxic to humans but despite the risk it has historically been used to enhance sexual function [88–90]. Its use dates back 2000 years in Chinese and African herbal remedies by grinding the beetle to a powder before dissolution in a solvent for ingestion [89]. When ingested, urethral irritation occurs which may cause priapism in both males and females [88, 90]. This occurs by inhibition of phosphodiesterase and protein phosphatase activity by cantharidin, which stimulates beta-receptors, inducing vascular inflammation of the genitourinary tract [2, 88]. This can lead to renal dysfunction which is common and related to acute tubular necrosis and glomerular destruction [88, 91]. Mucosal erosion and hemorrhage are seen in the upper gastrointestinal tract. Symptoms of can-

tharidin poisoning include abdominal pain, burning of the mouth, dysphagia, nausea, hematemesis, hematuria and dysuria [91]. Priapism, seizures, and cardiac abnormalities are less commonly seen [88]. When ingested by humans, the LD₅₀ is around 0.5 mg/kg, with a dose of as little as 10 mg being potentially fatal [92] although some have survived doses of up to 75 mg with treatment [87]. Tagwireyi et al. reported the intake of blister beetles in a 4-year-old girl and a 7-year-old boy in Zimbabwe [89]. They had collected some beetles, roasted them and ingested an unspecified number. After ingestion, both children had vomited and were lethargic when the parents found them. The brother died shortly after admission to hospital. The sister presented with many of the classical signs and symptoms of cantharidin poisoning including hematuria and abdominal pains. Externally, cantharidin is a potent vesicant (blistering agent), exposure to which can cause severe chemical burns. Properly dosed and applied, the same properties have also been used therapeutically, for instance for treatment of skin conditions, such as molluscum contagiosum infection of the skin. Lesions heal without scarring [93, 94].

Horny goat weed (*Epimedium*) and combinations with other plant extracts

Epimedium, also known as barrenwort, bishop's hat, fairy wings, horny goat weed, or yin yang huo, is a genus of flowering plants in the family Berberidaceae. The species mostly used as a dietary supplement are *Epimedium grandiflorum* and *Epimedium sagittatum* and are sold online under various names including Vi-

aXtreme, which also contains undeclared sildenafil [95].

Epimedium contains icariin, a prenylated flavonol glycoside (chemical structure see ■ Fig. 10), which is a weak phosphodiesterase V inhibitor [96] in vitro. Furthermore, it enhances the production of NO in the penis [97] as well as mimicking the effects of testosterone in rats [98]. The recommended doses of icariin for a quick effect usually range from 40 mg to 120 mg, but the use of very high doses (400–800 mg) has also been reported. Extracts from these plants are reputed to produce aphrodisiac effects and are used in traditional Chinese medicine to enhance erectile function [99].

No specific side effects are available for *Epimedium* intake in the literature. Some case reports are described: Partin and Pushkin [100] presented a case of a 66-year-old man with a history of cardiovascular disease who, after 2 weeks of consumption of herbal sexual enhancer containing horny goat weed, was admitted to hospital with a new onset tachyarrhythmia and hypomanic symptoms (sexual and verbal inappropriate behavior, irritable mood, and hyperverbal speech). The patient needed to undergo electric cardioversion and had to be treated with a therapy based on olanzapine and lorazepam [100]. Metz et al. reported a case of a painful vasculitic rash in a 77-year-old man, who had started a treatment with *Ginkgo biloba* and horny goat weed tablets 4 days before. The health status of the man improved after 3 days as an inpatient, with the suspension of the herbal supplement [101]. Further side effects have not been described in the literature. Qinna et al. described a herbal combination called Etana that is composed of five herbal extracts including *Panax quinquefolius* (ginseng), *Eurycoma longifolia* (Tongkat Ali), *Epimedium grandiflorum* (Horny goat weed), *Centella asiatica* (Gotu Kola) and flower pollen extracts. It led to a penile erection index in male rats comparable to sildenafil but did not show remarkable side effects and no acute and subacute toxicity [102].

Muira puama (potency wood)

Ptychopetalum olacoides Benth (Olanaceae), indigenous to the Ama-

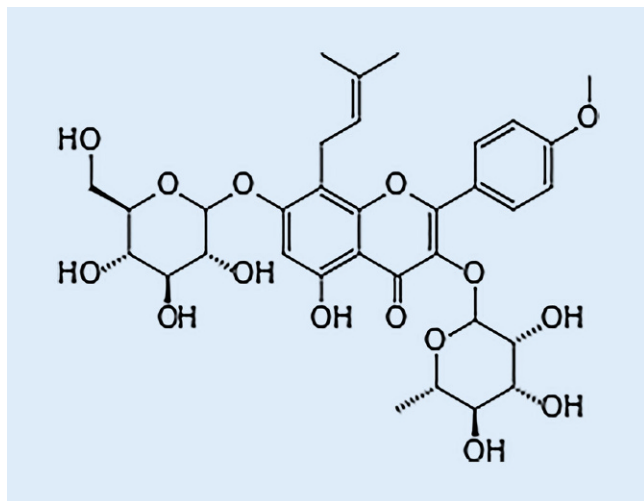


Fig. 10 ◀ Chemical structure of icariin

zon in Brazil and known as Marapuma/Muirapuama or potencywood [103–105] has a long history of use by Amazonian people [106]. It is commercially used as a sexual enhancer due to the popular belief that preparations from the roots can increase libido and the strength of the erection [104, 106, 107] and also for the intensification of sexual performance, as a body stimulant, nerve tonic and for the treatment of chronic degenerative conditions [105, 108, 109].

Phytochemical analysis revealed the presence of alkaloids, tannins, saponins, flavonoids and phenolic compounds [109]. Muirapuama was shown to increase the level of the cyclic adenosine monophosphate (cAMP) [110]. This could explain the demonstrated smooth muscle relaxant effects of the corpus cavernosum of rabbits and furthermore the prosexual effects for men. The observed relaxation had a rapid onset but only a short duration of action [12, 110]. Siqueira et al. examined the psychopharmacological effects of Muirapuama, reasonable by interacting with neurotransmitter systems and showed an excitatory action on the central nervous system which could support the proposed hypothesis of the interaction with dopaminergic and/or noradrenergic systems [109].

Data on safety and toxicity are very rare. There is just some information available from research on mixed formulation, containing other ingredients beside Muirapuama. One of these herbal

formulations is Herbal vX, which was assessed in 202 healthy women in a study conducted by Waynberg and Brewer in 2000 [107]. Only eight of these women reported adverse effects, which included sweats/headache/irritability, headache/thrush and stomach cramps. None of the mentioned adverse events were reported twice. Several women reported positive side effects as for example increasing energy [107].

Catuama® capsules contain a herbal extract of *Paullinia cupana* (Guarana), *Trichilia catigua* (Catuaba), *Ptychopetalum olacoides* (Muirapuama), and *Zingiber officinale* (Ginger) and are commercially available in Brazil and over the internet in other countries [103, 108]. Sexual stimulation is one of the attributed effects [108]. In a study by Oliveira et al. 25 mL Catuama® was given twice a day to 58 volunteers (25 mL Catuama® contains 0.875 mL *P. olacoides* fluid extract). No serious adverse events were reported. Five times moderate headache and two times pyrosis (once treatment required) were reported. Just once there was documented sleepiness for 1 week, vomiting, diarrhoea for 3 days and insomnia during the whole period of treatment [108].

Maca root

Maca (*Lepidium meyenii*) belongs to the Brassicaceae family and grows almost exclusively at high altitude in the Peruvian Central Andes. Its root has been used as food and in traditional medicine by the inhabitants of the Peruvian Cen-

tral Andes for approximately 2000 years [111–113]. They attribute the improvement of sexual function, increasing of sexual desire and enhancement of fertility to the dried hypocotyls [111, 114]. On the Internet, maca is one of the most cited plant extracts for the improvement of sexual desire (especially for women) [114]. The root contains different secondary plant compounds, mainly glucosinolate and imidazole alkaloids, fatty acids and macamide were proven [111].

There are only few publications which could be used for an evaluation of acute side effects related to Maca products with only small study collectives and some systematic flaws [111, 114]. Shin et al. made a systematic review about maca for improving sexual function. During their research they found four randomized controlled trials. All of these trials did not attempt to assess adverse effects of maca [114]. Gonzales-Arimborgo et al. showed that black maca and, in smaller proportions, red maca reduced hemoglobin levels in highlanders with abnormally high hemoglobin levels; neither variety of maca reduced hemoglobin levels in lowlanders. Black maca reduced blood glucose levels. Maca extracts did not show serious adverse effects [112]. Piacente et al. showed the existence of 1R,3S-1-methyl-1,2,3,4-tetrahydro- β -carboline-3-carboxylic acid (MTCA) in the extracts of maca. This has been suggested to be an inhibitor of the monoamine oxidase (MAO) enzyme and is a comutagenic or a precursor to mutagenic compounds [115].

Panax ginseng

Panax ginseng, a herb from the *Araliaceae* family within the *Panax* genus, is one of the most commonly used and best-selling herbal drugs worldwide [116–118]. Native to Korea and China, it has been used in eastern Asia for many centuries not only as food but also as a remedy due to its wide range of attributed health benefits [117–119]. Even today it is applied for many health conditions including improvement of the immune system, fatigue relief, blood circulation improvement and sexual dysfunction even if the evidence is encouraging but also questionable. Ginsenosides are the

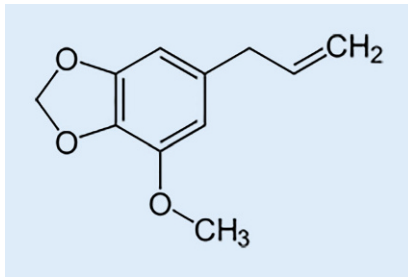


Fig. 11 ▲ Chemical structure of myristicin

unique and characteristic components of this plant, which belong to the group of saponins. The therapeutic action is believed to be based on this mixture of triterpene saponins [117, 119]. *P. ginseng* is prepared and manufactured in several different ways leading to a wide range of forms, for example fresh ginseng, white ginseng and red ginseng [116, 118]. Red ginseng, cultivated for 6 years and manufactured by steaming and then drying, has caused the highest aphrodisiac effects [12, 119]. Nitric oxide seems to be the mediator for the relaxation of the penile corpus cavernosum because it could be shown that ginsenosides promote the generation of NO in endothelial cells and perivascular nerves [119, 120].

The conclusion in a review by Thompson et al. [118] was that the risks of *P. ginseng* seem to be only very small. Clinical trials indicate that the incidence of adverse events by the treatment with monopreparations of ginseng is similar to that of a placebo [116, 118, 119]. The rarely documented adverse effects are mild (headache, sleep and gastrointestinal disorders) [118]. Among these commonly observed events were headache, sleep and gastrointestinal disorders. Combination products containing ginseng are more often related with serious adverse events. It should be noted that the other ingredients of this products may be the cause of adverse effects [116, 118]. Potential risk factors could be contamination and mislabeling of ginseng products, what could explain reported adverse events in case reports. Many investigations showed a problem with pesticide residues in herbal materials. Potential drug interactions have been described with phenelzine, warfarin and alcohol [116, 118].

Myristica fragrans (Nutmeg)

Nutmeg is derived from the seeds from the fruit of the *Myristica fragrans* tree, a bushy evergreen indigenous to India, Indonesia, and Sri Lanka [121]. The seeds are dried gradually in the sun over a period of 6–8 weeks. During this time the nutmeg shrinks away from its hard seed coat until the kernels rattle in their shells when shaken. The shell is then broken with a wooden club and the nutmegs are picked out. In low doses, nutmeg produces no noticeable physiological or neurological response, but in large doses, raw nutmeg has psychoactive effects deriving from anticholinergic-like hallucinogenic mechanisms attributed to myristicin (chemical structure see ■ Fig. 11) and elemicin [122, 123]. Historically, nutmeg was cultivated for use as an aphrodisiac and medicinal drug, with treatment ranging from pain reduction to appetite enhancement and nerve stimulation.

Myristicin, a monoamine oxidase inhibitor and psychoactive substance [122] can induce convulsions, palpitations, nausea, eventual dehydration, and generalized body pain. In overdose situations nutmeg can be associated with side effects such as excitedness, anxiety, confusion, headaches, nausea, dizziness, dry mouth, redness in eyes, and amnesia. Nutmeg poisoning is also reported to induce hallucinogenic effects, such as visual distortions and paranoia [124].

Ginkgo biloba

Ginkgo biloba is a living fossil tree, belonging to Ginkgoaceae family. The extracts obtained from Ginkgo leaves contain flavonoid glycosides (e.g., myricetin and quercetin) and terpenoids (e.g., ginkgolides and bilobalides) and have been used for centuries in traditional Chinese medicine, particularly for memory enhancement and blood flow improvement. These ingredients have been suggested to have the ability to inhibit monoamine oxidase (MAO) enzyme and the uptake of certain neurotransmitters such as noradrenaline and serotonin in the central nervous system [125]. Ginkgo extracts have been suggested to improve sexual function by improving the blood flow to the brain as well as

to the genital organs. These extracts have also been suggested to increase NO bioavailability, which may have positive impact on sexual function [126].

In reviews of the efficacy and safety of Ginkgo extracts in human subjects, no serious side effects were reported in clinical trials and side effects were similar to those seen in the placebo groups [127]. Ginkgo extract has been suggested to affect platelet aggregation, which could cause spontaneous bleeding in healthy individuals or patients treated with anti-coagulants such as warfarin and aspirin [128]. Moreover, there have been case reports of subdural hematoma or cerebral hemorrhage with headache, confusion, blurred vision, nausea, and vomiting after Ginkgo administration [129]. Some Ginkgo extracts may contain 4'-O-methylpyridoxine, which is a neurotoxin also known as ginkgotoxin. Ginkgotoxin poisoning is characterized by epileptic convulsions, vomiting, unconsciousness, and irritability, which may be fatal if not promptly treated [130].

Nymphaea caerulea

Nymphaea caerulea is a lotus flower belonging to Nymphaeaceae family. It is also known as blue water lily or Egyptian lotus and it is sold as a powder named blue lotus. The flower has been used by ancient Egyptians not only to enhance sex drive and to improve sexual performance, but also to stimulate blood flow and as an antiaging treatment. It has also recreational potential when added to wine or smoked, due to its psychoactive properties due to the psychoactive opioid aporphine (chemical structure see ■ Fig. 12) [131]. The flower may induce both narcotic and euphoric effects and, if taken in large amounts, may cause slightly hallucinogenic symptoms [132].

Saw palmetto

Saw palmetto (*Serenoa repens*, *Serenoa serrulata*) is one of the most common native palms in the United States of America and another natural aphrodisiac; the plant's berries are used to increase muscular power and libido [133, 134]. The liposterolic extract of the berries is a phytotherapeutic drug very frequently used by men for lower urinary tract symp-

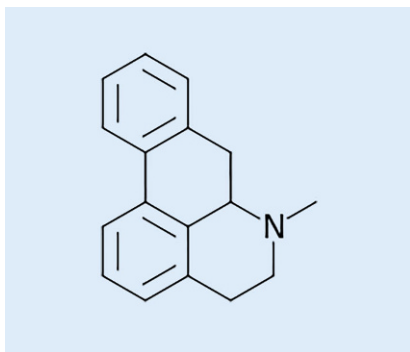


Fig. 12 ▲ Structure of aporphine

toms due to benign prostatic hyperplasia [133–135]. Saw palmetto is said to treat male impotence, sterility and reproductive problems, due to the reported increasing of blood flow to sexual organs and affection on the thyroid gland's regulation of sexual development [133].

There are two studies on the potential toxicity of saw palmetto including both symptomatic adverse effects as well as laboratory abnormalities [134, 135]. In a first study 225 men were randomized and saw palmetto extract (120 mg) or placebo was given twice a day over a period of 1 year [134]. In a second study, 369 patients were randomized to 320 mg, 640 mg and 960 mg of a saw palmetto extract or identical placebo daily [135]. The observation period was 18 months long. No serious toxicity of saw palmetto was observed in both clinical trials. The second trial found more non-serious adverse events; however, the difference was small and not significant. The most common adverse events were minor gastrointestinal symptoms, genitourinary problems, musculoskeletal complaints and upper respiratory tract infections [135]. Both trials found a significant difference in abnormal hemoglobin levels, but the differences were small [134, 135]. Another study led to the conclusion that this plant did not produce any significant effect on the liver [136]. Recent reviews have generally concluded that saw palmetto seems to be relatively safe; however, there are a few case reports suggesting that a serious idiosyncratic toxicity of saw palmetto may exist. The adverse effects included hepatotoxicity, coagulopathy, pancreatitis and intraoperative floppy-iris syndrome [134, 135].

Further herbal and animal products without known side effects

The following plants or their extracts are also used for the treatment of sexual dysfunction or for increasing libido. There are no studies on acute side effects and toxicity in men available.

The rhizome of the perennial herb *Alpinia calcarata* Roscoe (Zingiberaceae), cultivated in tropical countries including India, Sri Lanka and Malaysia, is used as an aphrodisiac in the Sri Lankan traditional Ayurveda medicine [137, 138]. A phytochemical analysis revealed the presence of polyphenols, tannins, flavonoids, steroid glycosides, alkaloids and essential oil [137, 138]. Ratnasooriya et al. showed prolonged ejaculation time, but also examined the risk of intoxication due to chronic use, but no treatment-related deaths or morbidity was observed in rats. Furthermore, no hepatic damage or renotoxicity was evident [139]. No studies on side effects in men are described in literature.

Extracts of *Anacyclus pyrethrum* (Spanish chamomile), which is a plant containing an alkylamide comparable to Maca, have anabolic activity in mice and also increase testosterone in the animal model [140]. No studies on side effects in men are described in literature.

Tuberous roots of *Chlorophytum borivilianum* (CB) from India (commonly known as safed musli) (Liliaceae) is an ayurvedic herb known as Vajikaran Rasayana, which is used for improving potency and alleviating sexual dysfunction [141, 142]. No studies on side effects in men are described in literature.

Crocus sativus L., commonly known as saffron, belongs to the family Iridaceae and is widely cultivated in Iran and other countries, including India and Greece. In traditional medicine, saffron is recommended as an aphrodisiac agent. The effects of saffron stigma extract on sexual behaviors in male rats are supposed to be due to two active constituents, crocin and safranal [143]. Safarinejad et al. conducted an open label, randomized, fixed-dose, crossover study comparing efficacy and safety of sildenafil citrate and saffron for treating ED in men naive to treatment. Side effects were not described; however,

treatment satisfaction was found to be very low in saffron patients [144].

Eurycoma longifolia (commonly called tongkat ali or pasak bumi) is a flowering plant in the family Simaroubaceae, native to Indonesia, Malaysia, and to a lesser extent, Thailand, Vietnam, and Laos. The root of the plant is boiled in water, and the water is consumed. It has gained notoriety as a symbol of male ego and strength by Malaysian men because it increases male virility and sexual prowess during sexual activities. No significant side effects of the treatment are obvious after the intake in clinical trials [145–147].

Fadogia agrestis is a plant from Nigeria which possesses significant aphrodisiac potential. The aqueous extract of *F. agrestis* stem increased the blood testosterone concentrations and this may be the mechanism responsible for its aphrodisiac effects and various masculine behaviors [148]. No studies on side effects are described in literature.

Kaempferia parviflora (Thai black ginger, Thai ginseng, Krachaidum) belongs to the family Zingiberaceae and is a native plant of Southeast Asia. It is traditionally used to enhance male sexual function; however, only few scientific data in support of this anecdote have been reported [149]. Phytochemicals of Krachaidum contain 2 major constituents: 5,7-dimethoxyflavone and 5,7,4'-trimethoxyflavone [150]. Saokaew et al. reviewed literature on clinical and toxicological effects of Krachaidum and found seven randomized controlled trials with the intake of *Kaempferia parviflora* extracts. Adverse events were not reported among all included studies [151].

Osthole is a naturally occurring bioactive coumarine found in medicinal plants such as *Cnidium monnieri* and *Angelica pubescens* whose extracts are used in traditional Chinese medicine to enhance sexual activity. Recommended doses are 50–200 mg per day for a long-term treatment and 400–500 mg for a quick effect [152]. *Osthole* was detected in 7 of 150 tested formulations at levels of 3–31 mg, alone (once) or together with DHEA (twice) [10]. The mechanisms of multiple pharmacological activities of *osthole*

are very likely related to the modulatory effect on cyclic adenosine monophosphate (cAMP) and cyclic guanosine monophosphate (cGMP) level, though some mechanisms remain unclear [153]. There has been no publication on side effects and toxicity of osthole in humans.

Mondia whitei is from the Periplocaceae family and has been used by many traditional medicine practitioners for the management of ED. It is used to increase libido, to prevent ejaculation and also for the management of low sperm count [154, 155]. No studies on side effects are described in literature.

Montanoa tomentosa (Zoapatle), a plant native to Mexico and Central America has an extensive ethnomedical history of use as a traditional remedy for sexual impairment [156]. It is suggested that the aqueous extract exerts its aphrodisiac properties by increasing sexual potency acting as an oxytocic agent. Landgren et al. [157] could show no side effects when administering the extract orally to six pregnant women.

Phoenix dactylifera (date palm) of the family Palmae is a native to North Africa and has been extensively cultivated on the arabian peninsula and the Persian Gulf. The date palm pollen (DPP) is used in the traditional medicine for male infertility [158]. No studies on side effects are described in literature.

Satureja khuzestanica Jamzad belongs to the family Lamiaceae and is an endemic plant that is widely distributed in the southern part of Iran. Its leaves are used as an aphrodisiac [159]. The drug was described to be safe and without adverse effects in men [160].

Spilanthes acmella is widely distributed in tropical and subtropical countries and mainly found in India, Sri Lanka, and South America where it is used for the treatment of various sexual problems especially for the improvement of sexual function in men [140, 161]. Phytochemical investigations of the ethanolic extract of *S. acmella* reported that it contains a certain amount of N-alkylamide [140, 161]. The main N-alkylamide of Maca, called macamide, has been shown to be effective in improvement of sexual behavior [140]. It thus stands to reason that alkylamides

have an effect on androgen production by mimicking the action of testosterone or stimulate the secretion of it and releasing NO [140, 161]. No acute toxicity of the ethanolic extract could be shown in animal studies [140, 162]. However, hexanic extract of this herb has been reported to trigger full tonic-clonic convulsions in rats [162]. No studies on side effects in men are described.

Terminalia catappa is a large tropical tree that belongs to the family Combretaceae with a significant aphrodisiac potential in rats. Ratnasooriya et al. observed that *T. catappa* seeds at dose of 1500 mg/kg or 3000 mg/kg, per oral for 7 days in rats had a marked improvement of aphrodisiac action and sexual vigor. No studies and side effects in men are described [163].

Turnera diffusa, also known as damiana, is a small shrub common in Mexico, Central and South America, and the Caribbean. Aphrodisiac and psychoactive properties have been ascribed for centuries to *Turnera* leaves [164]. The leaves are usually boiled to make an infusion with the aim of intensifying sexual sensations; moreover, when smoked, damiana leaves may produce similar effects to marijuana [165]. It has recently been found in products advertised on the Internet to increase sensuality, especially among women [166], as legal and safer alternative to cannabis and sold as mystical incenses [167]. Furthermore, Damiana is often used as the primary herbal product onto which synthetic cannabinoids are sprayed [168].

Velvet antler has been used in traditional Chinese medicine for centuries, mainly as an energizer and to treat joint pain, cardiovascular diseases, and impotence. The name refers to the velvety skin covering growing antlers of male elk, deer, and moose, which can be removed with no harm for the animal. The most relevant issue currently identified for elk or deer velvet antler consumption is related to an infectious neurodegenerative prion disease known as chronic wasting disease (CWS). The core symptoms of this encephalopathy are notable weight loss and behavioral abnormalities (e.g., progressive hyperexcitability) [169].

Conclusion

Forensic toxicologists should be aware of the potential toxicity of libido-enhancing products from the internet. Analytical targets for the detection of some of these products may be detected easily within routine workflows (e.g. sildenafil or GHB), some may be part of mass spectrometric screening libraries (e.g. yohimbine); however, most of the substances are not part of the commercially available screening libraries and may be overlooked.

Products are often sold over the counter and therefore casually used. Beneficial effects are controversial. The main products that could cause intensive side effects and intoxications are phosphodiesterase V inhibitors (declared or undeclared on package), yohimbine, steroid hormones, flibanserin, GHB, *Tribulus terrestris*, Bufo Toad, Spanish fly or Horny Goat Weed. All other described products used as aphrodisiacs show less potential side effects or no studies on side effects in man have been conducted.

Within the described case findings included unspecific symptoms like nausea and vomiting or dizziness. The specific symptoms disorientation, problems in vision and speaking, numbness, psychomotor problems, feeling not to be aware of senses, disinhibition, euphoria and anterograde amnesia, could only be attributed to GHB or its precursors which could explain all of these symptoms in the described case report. Amnesia and lapses of memory were not described and cannot be explained by the intake of other aphrodisiacs described here; however, there are no data in the literature regarding the combination of these aphrodisiacs with other centrally acting substances like ethanol.

A limitation of this study is that only publicly available publications and websites were monitored. Monitoring of private ways of discussion fora could yield more potential side effects of aphrodisiacs by users themselves. Furthermore, the combination of various ingredients can lead to unforeseeable increase of desired and adverse effects (for example in 1 study up to 33 ingredients were identi-

fied in 1 formulation [37] or combination with alcohol).

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Compliance with ethical guidelines

Conflict of interest. C. Hess and A. Boehmer declare that they have no competing interests.

Ethical standards. For this article no studies with human participants or animals were performed by any of the authors. All studies performed were in accordance with the ethical standards indicated in each case.

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